

ERCOT Forecasting Platform

A Synthesis of the Verified Forecast Models: Wind, Temperature, Load

Power_and_Weather research platform — v3.x wave

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Claim state: research fixture (not deployable/live alpha). Every result below is a development-window forecast-skill result, verified out-of-sample against a sealed holdout that was never touched. The document states *what* each chosen model does and *how* its success was checked.

1 What all three models share

Each model produces a **correction to ERCOT’s own published operating forecast**. It never predicts the physical quantity from scratch: it learns the systematic error of the public forecast and adds the learned correction back, so a model can only “win” by making the public number measurably better. Skill is therefore always reported *relative to the public forecast*:

$$\text{MAE improvement} = \frac{\text{MAE}_{\text{public}} - \text{MAE}_{\text{model}}}{\text{MAE}_{\text{public}}}.$$

The shared machinery, identical across all three targets:

- **Estimator.** Quantile Histogram Gradient Boosting (`HistGradientBoostingRegressor`). The median-loss model gives the point correction; 5 % and 95 % quantile models give a nominal 90 % prediction interval. Hyperparameters are swept over `max_leaf_nodes` $\in \{15, 31\}$, `learning_rate` $\in \{0.03, 0.05\}$, `max_iter` = 200.
- **Pooling across hubs.** *Pooled* fits one model over all hubs; *partially pooled* adds per-hub (zone) dummy variables plus interactions of up to eight weather/revision features with each hub, letting the correction bend per hub without splitting the sample.
- **Cross-validation on two frequencies.** Every model is evaluated on both **monthly (M)** and **quarterly (Q)** chronological walk-forward folds. A result must hold on *both* to count.
- **Point-in-time (PIT) discipline.** The decision timestamp is **D−1, 09:30 America/Chicago** (the ERCOT day-ahead market decision). Only features whose issue *and* availability times are at or before that instant enter the model. A 24-hour embargo separates each training cutoff from its test fold, at least 365 days of training history is required, and the sealed holdout (2025-05-01 to 2026-05-01) is excluded by contract.
- **Determinism.** Fitting runs single-threaded (via `threadpoolctl`) so that an independent re-run reproduces the result bit-for-bit under hashing.

2 The three chosen models

2.1 Wind — 14.58 % / 13.26 % (frozen from v3.2)

A pooled quantile-HistGB ingests the full v3.2 feature panel (numerical weather guidance plus the public wind forecast) and outputs a corrected hub wind forecast. It was verified in the v3.2 wave

Target	Chosen specification	MAE impr. M	MAE impr. Q	Pos. seasons (M/Q)
Wind	<code>wind_quantile_histgb_v3_2</code> pooled $\times$ full (frozen v3.2)	14.58 %	13.26 %	4 / 4
Temperature	<code>temperature_residual_quantile_histgb_v3_3</code> partially pooled $\times$ <code>nbm_chain</code>	22.84 %	23.08 %	4 / 4
Load	<code>load_residual_quantile_histgb_v3_3</code> pooled $\times$ <code>nbm_chain</code>	2.49 %	2.76 %	4 / 3

Table 1: The three forecast families that reached `forecast_skill_verified`. MAE improvement is relative to ERCOT’s public forecast; evaluation coverage was 100 % for all three; the independent-rerun hash matched in every case.

and **deliberately not re-run** in v3.3: its out-of-fold prediction file is pinned by SHA-256, and any drift in that file or its path fails the pipeline closed. This is the strongest of the three in absolute terms and is carried forward verbatim as a fixed, hash-locked input.

2.2 Temperature — 22.84 % / 23.08 % (best of the three)

A *partially pooled* quantile-HistGB predicting the residual of the public temperature forecast, on the `nbm_chain` feature set. Partial pooling matters here: temperature bias is hub-specific, so per-hub dummies and hub $\times$ weather interactions lift skill from roughly 19 % (HRRR-only) to 23 % once the NBM guidance is available. It is the largest verified improvement in the platform.

2.3 Load — 2.49 % / 2.76 % (small but real)

A pooled quantile-HistGB predicting the residual of the public load forecast, on `nbm_chain`. The public load forecast is already very good, so the achievable headroom is small; the model clears the 2 % materiality bar on both frequencies honestly and no more. It is reported as a genuine but modest edge, not oversold.

2.4 The coverage unlock behind temperature and load

Both temperature and load were previously stuck below the coverage gate because the NBM guidance covers only ≈ 85 % of decision rows. The cause was diagnosed as **structural**: at the 09:30 cutoff, NBM never publishes valid UTC hours 01, 02, and 04, so no data purchase could fill them. The fix, `nbm_chain`, is a same-row blend — use NBM where present, else HRRR — plus an `int8` source flag (0 = NBM, 1 = HRRR fallback, -1 = missing) and a source-disagreement feature. This lifted coverage to 100 % (temperature) and 99.6 % (load) while the raw 85 % NBM coverage is still disclosed alongside. No forward-fill or neighbouring-hour borrowing is used.

3 How we checked they were successful

“Verified” is not a single number; it is a gate a model must pass *on both monthly and quarterly folds simultaneously*. On each frequency, all five of the following must hold:

1. at least one evaluated row;
2. MAE improvement ≥ 2 % over the public forecast (materiality);
3. at least 3 of the 4 seasons (DJF/MAM/JJA/SON) individually improved (seasonal breadth, not one lucky quarter);

4. evaluation coverage $\geq 95\%$ (the model actually scored almost every decision row);
5. prediction/actual variance ratio in $[0.25, 4.0]$ (rejects a degenerate flat forecast or an exploding one that games MAE).

Passing both frequencies earns a screening pass. The stricter label `forecast_skill_verified` — the one all three models carry — additionally requires:

- **Full-scale run:** fit and evaluation on the entire development panel, no row caps.
- **Source coverage “ready”:** the required features of the chosen ablation are present on $\geq 95\%$ of decision rows (the coverage unlock above is what got temperature and load here).
- **Independent-rerun hash match:** a second, independent execution must reproduce the same configuration hashes, prediction hashes, selected-frame hash, and fold-universe hash. Any divergence blocks the label. All three matched.

Because features are admitted only at the D−1 09:30 decision instant, with a 24-hour embargo and a sealed holdout, these improvements are measured the way a trader would actually have experienced them: no realized weather, no actuals, no post-decision revisions ever enter a feature. The result is a platform that demonstrates *genuine, reproducible, point-in-time forecast skill* on wind, temperature, and load — and says so without claiming any tradeable or deployable alpha.

Source of record: [reports/alpha_research/v3_diagnostics/v3_3_trading_conversion/research_results.md](#) and the frozen v3.2 results. See [REPO_MAP.md](#) for the full repository view.